

MATH 1075 - Section 2.8 and 9.1 Worksheet

10/01/2026

1 Solving Inequalities

Solve following inequalities and write your answer in both set notation and interval notation.

1. $y - 7 > 6$

2. $5r + 4 \geq 4r - 1$

3. $-3c > 6$

4. $8 - 6(x - 3) > -4x + 12$

5. $\frac{1}{3}w - \frac{1}{2} \leq \frac{5}{6}w + \frac{1}{2}$

6. $-15 < 3x + 6 \leq 12$

7. $5 \leq -3x - 2 < 8$

2 Intersections and Unions of Sets

1. Given two sets $P = \{a, b, c, d, e, f, g, h, i\}$ and $Q = \{a, e, i, o, u\}$, find

a) $P \cup Q$

b) $P \cap Q$

2. Given sets $A = \{x \mid x < -4\}$, $B = \{x \mid x > 2\}$ and $C = \{x \mid 0 \leq x \leq 5\}$, find

a) $A \cup B$

b) $A \cap B$

c) $A \cup C$

d) $A \cap C$

e) $B \cup C$

f) $B \cap C$